Hani Sabaie, M.Sc.

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Profiles: Google Scholar | ORCID | GitHub | LinkedIn | ResearchGate | Web of Science

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Education

Master of Science in Human Genetics | **Tabriz University of Medical Sciences** | Tabriz, Iran **2018–2021**
GPA: 4/4
Tuition-Free Master's Degree Program
Thesis: Analysis of *BCAS4/hsa-miR-185-5p/SHISA7* Competing Endogenous RNA Axis in Late-onset Alzheimer's Disease
Supervisor: Dr. Maryam Rezazadeh

Bachelor of Science in Biology-Animal Sciences | **University of Guilan** | Rasht, Iran **2012–2017**
GPA: 3.37/4
Tuition-Free Bachelor's Degree Program
Project: Genome Evolution
Mentor: Dr. Shohreh Siyam

Research Interests

- Statistical and Computational Genomics
- Structural Variation and Variant-to-Function Mapping
- Functional Genomics and Gene Regulation
- Clinical Transcriptomics and RNA Sequencing
- Single-Cell Multiomics Analysis

Research Experience

Research Assistant | **Cellular and Molecular Endocrine Research Center** | Tehran, Iran **2025–Present**
Principal Investigator: Dr. Mahdi Akbarzadeh
Primary duties:
- Perform integrative single-nucleus multiomics analyses to investigate cell-type-specific regulatory mechanisms underlying complex human diseases.
- Apply high-dimensional weighted gene co-expression network analysis (hdWGCNA) to single-cell transcriptomic data to identify disease-relevant gene modules and hub genes.
- Conduct Mendelian randomization, colocalization, and fine-mapping analyses to infer causal relationships between genetically regulated gene expression and disease susceptibility.
- Co-authored a review of computational methods for CNV detection and disease association, with emphasis on methodological advances and genomic medicine.
- Co-authored a comprehensive review of diagnostic transcriptomics in rare genetic disease, focusing on RNA-seq-based variant interpretation, aberrant splicing, computational outlier detection, and clinical implementation.

Research Assistant | **Tabriz University of Medical Sciences** | Tabriz, Iran **2024–Present**
Principal Investigator: Dr. Maryam Rezazadeh

Primary duties:

- Use systems biology, RNA-seq transcriptomic analysis, and weighted gene co-expression network analysis (WGCNA) to identify gene modules, hub genes, and regulatory networks associated with neurological conditions.
- Detect and interpret variants from exome sequencing data.
- Assist the principal investigator with manuscript and grant proposal preparation.
- Mentor medical, master's, and Ph.D. students in research activities.

Researcher | **Directorate of Health, Rescue and Treatment** | Tehran, Iran **2022–2023**

Project:

- Conducted a research project under the Iranian National Elite Foundation program in lieu of mandatory military service.
- Integrated bioinformatic and experimental approaches to identify peripheral blood-based biomarkers for major depressive disorder.

Student Research Assistant | **Tabriz University of Medical Sciences** | Tabriz, Iran **2019–2021**

Principal Investigator: Dr. Maryam Rezazadeh

Main Projects:

- Identified and examined ceRNA regulatory axes in neurogenetic disorders by reanalyzing publicly available microarray datasets and applying complementary molecular methods.
- Assessed the expression of key genes in neurogenetic conditions using PCR, RT-PCR, and agarose gel electrophoresis.
- Conducted systematic and scoping reviews of genetic aspects of human diseases, with an emphasis on neurogenetic conditions.

Selected Peer-Reviewed Publications

1. Saed A, Akbarzadeh M, Gohari F, Asadrouh N, Jahromizadeh AM, **Sabaie H**, Zarkesh M, Hedayati M, Azizi F, Daneshpour MS. Serum ferritin and delirium risk: an integrative genomic analysis of causal inference and multi-tissue regulatory signals. **Human Genomics**. 2026. doi:10.1186/s40246-026-00972-5

2. **Sabaie H**, Taghavi Rad A, Shabestari M, Habibi D, Saadattalab T, Seddiq S, Saeidian AH, Zahedi AS, Sanoie M, Vahidnezhad H, Zarkesh M, Foroutani L, Hakonarson H, Azizi F, Hedayati M, Daneshpour MS, Akbarzadeh M. Mitochondrial DNA Copy Number as a Hidden Player in the Progression of Multiple Sclerosis: A Bidirectional Two-Sample Mendelian Randomization Study. **Molecular Neurobiology**. 2025;62(9):11643-53. doi:10.1007/s12035-025-04980-9

3. **Sabaie H**, Taghavi Rad A, Shabestari M, Seddiq S, Saadattalab T, Habibi D, Saeidian AH, Abbasi M, Mirtavoos-Mahyari H. Deciphering the bidirectional impact of leukocyte telomere length on multiple sclerosis progression: A Mendelian randomization study. **Multiple Sclerosis and Related Disorders**. 2025;94:106277. doi:10.1016/j.msard.2025.106277

4. Shoaran M, **Sabaie H**, Mostafavi M, Rezazadeh M. A comprehensive review of the applications of RNA sequencing in celiac disease research. **Gene**. 2024;927:148681. doi:10.1016/j.gene.2024.148681

5. **Sabaie H**, Gharesouran J, Asadi MR, Farhang S, Ahangar NK, Brand S, Arsang-Jang S, Dastar S, Taheri M, Rezazadeh M. Downregulation of miR-185 is a common pathogenic event in 22q11.2 deletion syndrome-related and idiopathic schizophrenia. **Metabolic Brain Disease**. 2022;37(4):1175-84. doi:10.1007%2Fs11011-022-00918-5

6. **Sabaie H**, Moghaddam MM, Moghaddam MM, Ahangar NK, Asadi MR, Hussen BM, Taheri M, Rezazadeh M. Bioinformatics analysis of long non-coding RNA-associated competing endogenous RNA network in schizophrenia. **Scientific Reports**. 2021;11(1). doi:10.1038%2Fs41598-021-03993-3

Complete list of publications available on [Google Scholar]

Teaching Experience

Teaching Assistant | **Tabriz University of Medical Sciences** | Tabriz, Iran **2021**

- Delivered weekly lectures on microarray data analysis.

Skills

Programming and Computational Skills:

- **R:** Proficient with statistical modeling, data analysis, visualization, and reproducible research.
- **Python:** Experienced with data analysis and scientific computing for genomic applications.
- **Linux/Unix and Bash:** Skilled with command-line workflows and pipeline automation.
- **Git/GitHub:** Experienced with version control and collaborative development.
- **Markdown/R Markdown:** Proficient with reproducible documentation and reporting.

Statistical and Computational Genomics:

- Skilled with two-sample and summary-data-based Mendelian randomization, colocalization and statistical fine-mapping (TwoSampleMR, SMR/HEIDI, MR-PRESSO, GCTA-COJO, coloc, and SuSiE).
- Experienced with Comprehensive Meta-Analysis (CMA).

Functional Genomics:

- **Single-Cell Multiomics:** Experienced with preprocessing, normalization, multimodal integration, clustering, trajectory inference, network analysis, motif activity, and cell–cell communication analysis (Drop-seq pipeline, Seurat, decontX, scDblFinder, Signac, Monocle, hdWGCNA, chromVAR, and CellChat).
- **Bulk RNA-seq:** Experienced with end-to-end workflows including QC, alignment, quantification, and differential expression (FastQC, Trimmomatic, HISAT2, STAR, featureCounts, HTSeq, StringTie, Kallisto, Salmon, edgeR, DESeq2, Ballgown, Sleuth, Picard, and GATK).
- **Co-expression and network analysis:** Skilled with co-expression, interaction, and ceRNA regulatory network analysis (Hmisc, psych, corrplot, bcdstats, LncBase, lncSNP2, miRTarBase, miRWalk, miRDB, ENCORI, HMDD, Cytoscape, MCODE, CytoHubba, STRING, MIST, GeneMANIA, Enrichr, FuncPred, g, ImmPort, WGCNA, and hdWGCNA).
- **Microarray Analysis:** Proficient with preprocessing, normalization, differential expression analysis (limma, affy, GEOquery, GEO2R, pheatmap, and EnhancedVolcano).
- **Genomic Variant Analysis:** Skilled with variant calling, annotation, and interpretation (FastQC, Trimmomatic, BWA, GATK, HaplotypeCaller, ANNOVAR, ExAC, gnomAD, 1000 Genomes, CADD, ClinVar, SIFT, PolyPhen, LRT, MutationTaster, FATHMM, PROVEAN, MutationAssessor, MCAP, MetaLR, Franklin, VarSome, CLC Genomics Workbench, ExomeDepth, ClinGen, and ACMG/AMP guidelines).

Experimental and Molecular Biology Skills:

- **Molecular Genetics Techniques:** Skilled with DNA extraction, RNA isolation, NanoDrop spectrophotometry, agarose gel electrophoresis, cDNA synthesis, PCR, RT-qPCR, and cell culture.
- **Primer Design and Reference Gene Validation:** Proficient with Primer3, Primer-BLAST, Oligo7, Gene Runner, NormFinder, and geNorm.
- **RT-qPCR Data Analysis:** Proficient with relative expression analysis and visualization (R, GraphPad Prism, SPSS, and GenEx).

Other:

- **Systematic Literature Search and Evidence Synthesis:** Proficient with PubMed, Embase, Scopus, Web of Science, Cochrane, ProQuest, and Google Scholar.
- **Reference Management:** Proficient with Zotero, EndNote, and ResearchRabbit.

- **Scientific Visualization:** Proficient with BioRender, Mind the Graph, and PowerPoint.

Conference Presentations

Poster Presentation:

- **Sabaie H**, Asadi MR, Salkhordeh Z, Rezazadeh M. Investigation of competing endogenous RNA regulation of BCAS4 and SHISA7 in tau pathology in Alzheimer's disease. **The first international conference and the tenth national bioinformatics conference of Iran**, 22-24 February 2022 in Kish Island, Iran.

Selected Computational Training

- **Comprehensive Training Program in Statistical Data Analysis, R, Python, and Machine Learning** | Shahid Beheshti University of Medical Sciences | Tehran, Iran (online) **2026**
- **Summary-data-based Mendelian Randomization** | Shahid Beheshti University of Medical Sciences | Tehran, Iran (online) **2025**
- **Mendelian Randomization** | Shahid Beheshti University of Medical Sciences | Tehran, Iran (online) **2024**
- **R Advanced Workshop for Data Science** | Shahid Beheshti University of Medical Sciences | Tehran, Iran (online) **2024**
- **R Preliminary Workshop for Data Science** | Shahid Beheshti University of Medical Sciences | Tehran, Iran (online) **2024**
- **Linux Essentials** | Shahid Beheshti University of Medical Sciences | Tehran, Iran (online) **2024**
- **R Programming** | Tabriz University of Medical Sciences | Tabriz, Iran **2020**
- **Basic Bioinformatics** | Tabriz University of Medical Sciences | Tabriz, Iran **2020**
- **Biostatistics** | Tabriz University of Medical Sciences | Tabriz, Iran **2020**

Honors and Awards

- **Selected Researcher** | Research Week, Clinical Research Development Unit, Tabriz Valiasr Hospital **2023**

Science Community Service

- **Reviewed manuscripts for international journals**, including:
  Scientific Reports; *Heliyon*; *PLOS ONE*; *Medicine*; *Aging*; *Annals of Human Biology*; *BMC Neurology*; *BMC Medical Genomics*; *Molecular Neurobiology*; *Molecular Medicine*; *Molecular Therapy – Nucleic Acids*; *Biomedicine & Pharmacotherapy*; *International Immunopharmacology*; *Functional & Integrative Genomics*; *Gene*; *Cell Cycle*; *Multiple Sclerosis and Related Disorders*; *Journal of Diabetes Investigation*; *Oncology Letters*; *Discover Oncology*; *Current Eye Research*; *Bioengineered*; *Iranian Journal of Allergy, Asthma and Immunology*; *Minerva Cardiology and Angiology*.

Volunteer Academic Service

- Mentored medical and graduate students in research design, practical implementation, and data analysis. **2022–Present**
- Advised candidates preparing for the Iranian national entrance examination for M.Sc. programs in Human Genetics. **2018–2021**

Interests and Activities

- Running
- Travel

- Reading in Philosophy and History

References

- **Maryam Rezazadeh, Ph.D.** (M.Sc. Thesis Supervisor)
  Associate Professor
  Department of Medical Genetics, Faculty of Medicine, Tabriz University of Medical Sciences, Tabriz, Iran.
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- **Mahdi Akbarzadeh, Ph.D.** (Principal Investigator)
  Assistant Professor
  Cellular and Molecular Endocrine Research Center, Research Institute for Endocrine Molecular Biology, Research Institute for Endocrine Sciences, Shahid Beheshti University of Medical Sciences, Tehran, Iran.
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  Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX, USA.
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  Professor and Director
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  Associate Professor
  Department of Pediatrics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA.
  vahidnezhh@chop.edu, Google Scholar
- **Leila Youssefian, Ph.D.** (Research Collaborator)
  Cytogenetics Associate Director
  Department of Pathology, Cytogenetics Laboratory, City of Hope National Medical Center, Duarte, CA, USA.
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